

Roll No.

Total Pages : 3

CMCS/M-22

24544

CYBER SECURITY AND BLOCKCHAIN

TECHNOLOGY

Paper–MS-20-43 (iii)

Time Allowed : 3 Hours]

[Maximum Marks : 75

Note : Attempt **five** questions in all, selecting **one** question from each Unit. Question No. **1** is compulsory. All questions carry equal marks.

Compulsory Question

1. Answer the following questions in brief :

- (a) What is Botnet?
- (b) What is keylogging? How does it differ from spyware?
- (c) How does steganography ensure security?
- (d) What is Digital Forensics? What steps are followed in digital forensics Life Cycle?
- (e) What are public blockchains and what are its advantages?
- (f) What is the need of consensus in Bitcoin network?

UNIT-I

2. (a) Give an introduction to the term cybercrime. What are the various categories and types of Cybercrimes?

- (b) What is information security and what can be the threats to information security? What are the goals of information/ cyber security and How can you protect yourself from cyber attacks?

3. Answer the following questions in brief :

- (a) How Proxy Servers and Anonymizers are used for cyber attacks.
- (b) List the types of viruses and explain any one of them. What is the difference between virus and worm.
- (c) What is the difference between SQL Injection attack and DOS attack?

UNIT-II

4. (a) What is Phishing and what are its methods and techniques?
- (b) What is identity theft? What are the common forms of identity theft?

5. Answer the following questions in brief :

- (a) What are the goals of firewalls and packet filters?
- (b) What is Network Address Translation and what are its advantages?
- (c) What is Virtual Private Network? Sketch a diagram to depict its layout.

UNIT-III

6. What is Blockchain and what is a Bitcoin? What is the need and applications of Blockchain?

7. Describe the following in the context of Blockchain :

- (a) Cryptographic Hash function.
- (b) Public Key Cryptography.
- (c) Digital Signature.

UNIT-IV

8. (a) How does Bitcoin handle double spending problem?
(b) How is a new block of Bitcoin transaction created?
How do Bitcoin transactions work?
9. What is meant by Bitcoin mining and Proof of Work (PoW)?
How is Hashcash used as a Bitcoin mining function?